

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Apache Spark, Python, Airflow, SQL, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered **Python** pipelines orchestrated by **Airflow DAGs** to process **50M+ daily transactions** from **Delta Lake**, feeding downstream **ML models** and analytics on **petabyte-scale** financial data.
- Optimized **Spark** jobs by migrating legacy code to **Spark-native** APIs and tuning **AQE/cluster memory**, solving resource bottlenecks and reducing **latency** by **40%**.
- Developing **multi-region** data pipelines with built-in **fault tolerance**, **idempotent** re-runs, and automated **failover** for critical banking workloads.

Software Engineer Intern | *TypeScript/Node.js, AWS, Playwright, OpenAI*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Engineered real-time **ETL** pipelines on **AWS (SQS/S3)** with **TypeScript/Next.js** + **Node** dashboards; owned design→deploy→diagnostics/**DevOps** (**CloudWatch**) at **thousands/day** scale.
- Accelerated content prep by ingesting brand docs with an **NLP** step, caching embeddings, and auto-injecting channel-specific context while enforcing **API guardrails**.
- Raised scheduler reliability to **99.9%** by moving generation to **OpenAI Responses API**, integrating **Playwright E2E** gates into **CI/CD**, hardening auth/session, and adding **retry/backoff** plus **CloudWatch alerting**.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led development of an **AI-driven** VR clinical simulation in **Unreal (C++)**, collaborating with UBC's School of Nursing to train nurse practitioners.
- Designed a novel **multithreaded supervisory architecture (C++/Python)** that monitors AI patient behavior and enforces clinical protocol adherence.
- Evaluated system performance, achieving **40%** latency reduction and **90%** inference cost savings through pipeline optimizations.

PROJECTS

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Building a unified freelancer workspace that bridges client intake, planning, proposals, and delivery.
- Shipped an email intake console that triages threads, captures scope, and renders versioned proposal PDFs with annotations via a **Vision Model** that translates intent for precise edits.
- Built a **Python** evaluator→revisor **agent** loop with rubric-based **benchmarking** (score + revision), improving response quality via measured iterations.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth*

Hack The Coast 2026

- Trained an **XGBoost** **ranker** for a TikTok-style investor feed, scoring startups on **6 features** (stage affinity, text similarity, tag overlap, popularity) with **sigmoid scoring**.
- Built a **supervised training pipeline** ingesting like/commit positives and sampled negatives from **DynamoDB**, with online enrichment from impression and watch-milestone **event streams**.
- Implemented the **ML feature layer**: investor affinity contexts, token-level text matching, and popularity normalization.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Java, Python, SQL, C/C++, Swift, HTML/CSS, React Native

Platforms & Tools: Next.js, Node.js/Express, Playwright; AWS (Amplify, Lambda, DynamoDB, S3, SQS, CloudWatch); OpenAI API; Meta/LinkedIn/X/Canva APIs; Git, Docker, Yarn, Tailwind CSS, RDBMS